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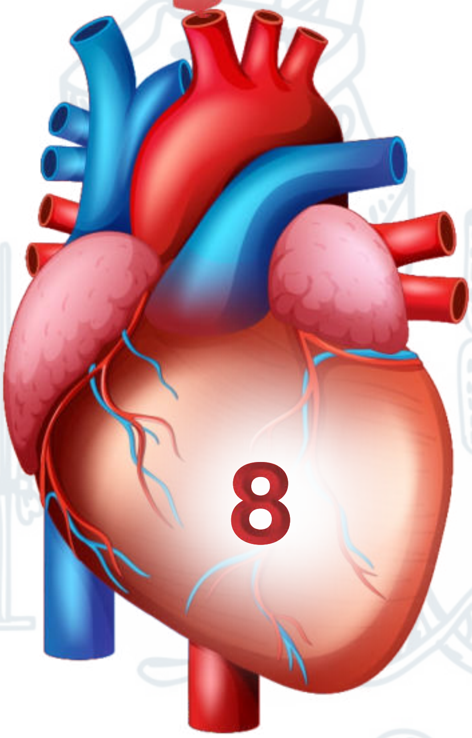
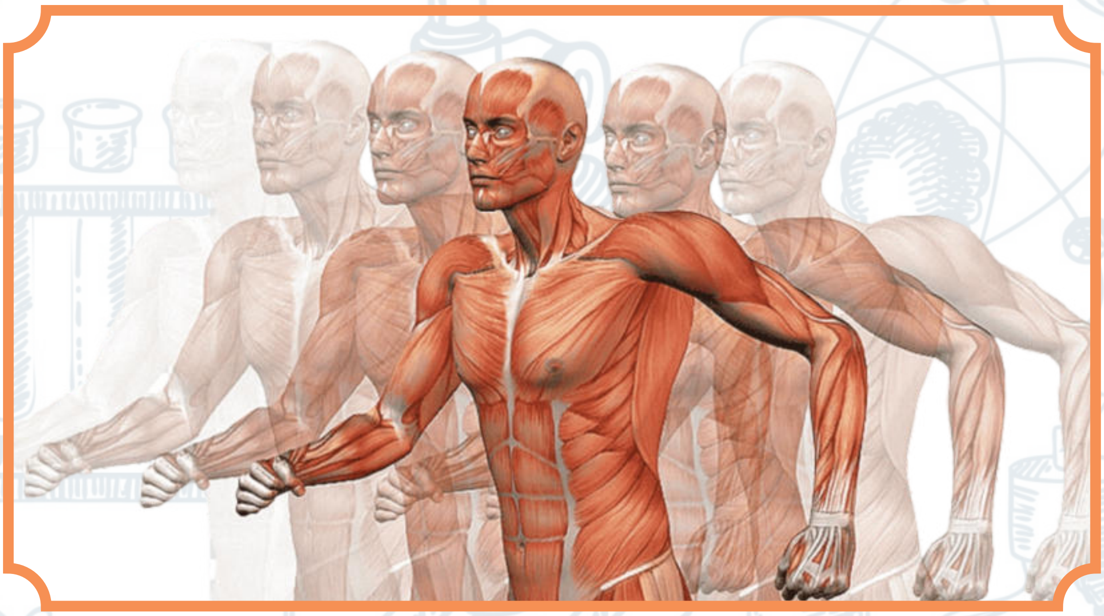


DIVERSITY AMONG PLANTS



THE EVOLUTIONARY ORIGIN OF PLANTS

In the beginning the plants were restricted only to aquatic conditions.
· The migration started towards land nearly 400 million years ago.
• · Many important features of land plants also appear in a variety of protists, primarily green algae. For example,
• Plants are multicellular, eukaryotic, photosynthetic, autotrophs, as
• are brown, red and certain green algae.
• Plants have cell wall made of cellulose. Likewise green algae,
• dinoflagellates and brown algae have also cellulose in their cell walls

Diagnostic Features of Plants

• Plants are multicellular eukaryotes with well-developed tissue and have autotrophic nutrition.
• Plants are well protected from being dried up in air by their cuticle,formed from a waxy
substance called cutin.
• The plant body has root, stems and leaves having vascular tissue xylem, phloem and cellulose rich cell walls.
• Plants show alternation of generation.
• It consists of the sporophyte the diploid generation that produces haploid spores by meiosis.
• Spores develop into a haploid gametophyte generation.
• The gametophyte produces gametes that unite to form a diploid zygote.
• The plants are oogamous, the gametes are eggs and sperms.

NON-VASCULAR PLANTS

General Characteristics of Bryophytes

1. Introduction
• The bryophyta is a group of plants comprising of liverworts, hornworts and mosses.
• · Bryophytes are typically quite small and a few exceed 2 centimetres in length.
• They generally require a moist environment for active growth and reproduction, but some
bryophytes tolerate dry areas.
• The gametophytes of bryophytes are green and manufacture their own food.
• · They are relatively large and prominent as compared to sporophytes.
• Some of their sporophytes are completely enclosed within gametophyte tissue, others that are not enclosed; turn brownish or straw coloured at maturity.

3) General Characteristics

The main features of bryophytes are:
• **I.** They lack specialized vascular tissues.
• **II.** Multicellular sex organs produce embryo. Spreading the Light
• **III.** Sporophytes are always smaller and obtain their food from the gametophyte.
• **IV.** Their life cycles involve alternation of generation.
• **V.** Bryophytes are also called amphibious plants because they need water for
development, existence and reproduction.

Diversity among Plants

Plants are multicellular eukaryotes with Photosynthetic nutrition. Cell typically have cellulose wall, sap vacuole, plastids and several photosynthetic pigments which always include chlorophyll-a. They have tendency to develop embryo and have heteromorphic alternation, of generation. Sexual and asexual both types of reproduction are reported in plants.

THE EVOLUTIONARY RELATION IN PLANTS

THE EVOLUTIONARY REDA earth about 3.5 billion years ago. The earliest plants have become the first organisms adapted for life on land. In the beginning the plants were restricted only to aquatic habitat but gradually they adapted to terrestrial habitat.

The evolutionary history of the plants kingdom is a story of adaptation to changing terrestrial condition.

There are wide variety of plants species exists on earth. They live in every type of habitat .They have colonized great areas of the earth surface in practically all sorts of soil and climatic conditions. They show enormous diversity in size and form.
Cuticular covering of the plant body, protective layer around the sex organs, evolution of an internal transport system of xylem and Phloem, specialised organs, such as leaves and roots and woody tissue allowed vascular plants to attain greater body size than smaller non-vascular plants, which rely on simple diffusion for transport of water and nutrients are the important characteristics in the transition of plant from water to land. Transport of male gametes in pollen to the female reproductive organ. The pollen grain dispersed by wind or animals rather than by water.

Although the earliest vascular plants were relatively small and probably, confined to swampy areas, after that a rapid diversification of plants occurred gradually.
"Plants may be defined as multicellular eukaryotes that are photosynthetic autotrophs (chlorophyllous) with cell wall primarily made up of cellulose , exhibiting heteromorphic alternation of generation and zygote retained to become develops into embryo".

General Characterstics of Plants:

- · Plants are multicellular eukaryotes.
- · They are Photosynthetic autotrophs and contain photosynthetic pigments specially chlorophyll, which enable the plants to convert light energy into food (chemical energy).
- · Rigid cell wall of plant cell primarily made up of cellulose
- · Life cycle of show heteromorphic alternation of generatiose.
- Life cycle of sed and develops into embryo

Plant cuticle, composed of insoluble lipid polymers and waxes filters a substant diffusion of water to the drier atmass be barries to prevent diffusion of water to the drier atmosphere and waxes filters a substantail component of the drier atmosphere and also Some others observent of UV radiation Some others characteristics in water to the drier atmosphere Some others characteristics included are:

plants produce some secondary products like sporopollenin and lignin. Sporopollenin from attack by grazing composed chiefly of carotenoids . It protects them from attack by grazing organism and microganism and microganism . It also make spores of land Plants tough and flexible and resible and resistants . It biodegradation. Lignin is important to provide strength to upright to upright terrestrial

In terrestrial environment gases enter through small pores in the plant surface. Stomata are present in modern hornworts, mosses stomata and lenticel and vascular are present in modern hornworts, mosses stomata as Rhynia. as Rhynia.

Vascular supply and lignin-Transport support:

The vascular transport of plants consists of xylem and Phloem for the transport of water and mineral ions and for the transport of sugar respectively. Another product Lignin provide strength to upright terrestrial tissues.

Alternation of generation and sexual reproduction:

Alternation of generation is the sexual reproduction: ucing generation alternate with sexual of life cycle where asexually reproducing generation alternate with sexually reproducing generation. It is an adaptation There are two alternating generation in plants life cycle. The gametophyte generation which is haploid produces male and female gametes by mitosis. Male and female gametes fuse to form zygote which grows into the diploid sporophyte generation. The diploid sporophyte produces spores by meiosis which are dispersed and germinate into new haploid between been in land plants, haploid and diploid generations are markedly heteromorphic, these
gametophyte and sporophyte differ in size and shape.

NON-VASCULAR AND VASCULAR PLANTS Non-Vascular Plants:

The non-vascular plants are the group of plants without specialized conducting tissues (xylem and phloem) for transporting water and nutrients. The gametophyte is the dominant stage of life cycle in these plants.

They need moist habitat to reproduce because flagellated sperm must swim through water drops, e.g. Liverwort, hornwort and mosses.

Bryophytes:

Bryophytes are the simplest non vascular land plants, first appeared before 450 million years ago. In bryophytes, strengthening and conducting tissue absent or poorly developed. It Includes Liver worts (Class Hepaticae), Horn worts (Class Anthocerotae) and mosses (Class Musci)

Bryophytes are small photosynthetic free-living gametophytes that lack vascular tissue in Bryophytes are small photosynthetic free-living gamestophyte and is dependent on it for Sexual reproduction requires water for sperin to swint on is dependent on it in an archegonium. The sporophyte grows utrition nutrition.

General Characterstics:

These are non-vascular plants, where xylem and phloem are absent.

i. Plants showing alternation of generation with dominam gametophytes having amphibious nature

ii. Gametophytes having aniphibious, chlorophyllous, photosyntheile Gametophytes are the conspitually or differentiated in rhizoida, Pseudo-stem and pseudo leaves.

iii• Male gametes (Antherozoid or spermatozoids) are produced in 111. antheridium (male gametangia, female gametangia). while female gametes (non-motile egg cell) is produced in archegonium.

iv• Sporophytes are semi-parasite on gametophytes having a body iv. differentiated into foot, seta and capsule. Spores are produced by the sporophytes from spore mother cell in capsule by meiosis.

v• They live mainly in damp, shady places.

vi. Bryophytes are amphibious in nature, need water to reproduce. Their sperms are flagellated and swim from antheridium to archegonium to fertilize the egg when archegonia mature and produce smell. The male gametes mature earlier than female gametes this property is called protoandry.

vii• Bryophytes lack the lignin fortified tissues required to support tall plants on land.

Life cycle of moss:

Life cycle of Bryophyte (Moss)

All bryophytes show heteromorphic alternation of generation with gametophytes as dominant generation including Funaria hygromothi will moss). The gametophyte is hapliod consisting of rhizoids, pseudostem and pseudo leaves.

The sex organs called antheridia (male) and archegonia (female) develop at the tips of pseudo stem which are always dioecious having either male of female sex organs. There is protoandry because antheridia mature earlier and liberate their anthrozoids, which start swimming with the help of their flagella in dew or rainwater . When archegonia mature, they have single ovum in the venter and few neck canal cells in the neck. Swimming sperins are attracted by scent of sugar cane secreted by mature (2n) oospore (zygote). This is retained with the ovum to form diploid embryo (2n). This embryo undergo repeated within archegonium and form and sporogonium (a sporophyte) which is diploid mitotic-divisions to form an sporogonium (a sporophyte) which is diploid. It consists of foot, seta and capsule. Within capsule spore mother cells are present. Each spore mother cell divides by meiosis to form four haploid spores. Each spore whendrop at proper substratum germinates into a filamentous body called protonema which develop into gametophyte,

Adaptations to land habitat:

Adaptations to land habitat: The first evidence that plants had invaded the land plants and animals The first evidence that plants had invaded the land plants and animals found in fossils. All the biologists agree that the land plants have been developing found in fossils. All the biologists agree that the land plants have been aevolved from aquatic ancestors. Life of aquatic organisms is an easy life. Water is necessary for the growth of new structures. new Life of aquatic organisms is an easy life. Carbon containing compounds, so essentail for autotrophs . Carbon containing compounds, in turn provide continuously supply of oxygen for all living organism in the sea.The temperature in the seas does not fluctuate as much as temperature on land Hence, the aquatic environment is more uniform .

When plants invaded the land from water, absorbing carbon dioxide from the atmosphere for photosynthesis.

To solve these problems, land invading plants adopted themselves first to amphibious-habitat and later developed a complete terrestrial form of life. The amphibious form of land plants includes all the bryophytes. We will consider the following adaptive characters exhibited by them.

- Rhizoids for water absorption
- Conservation of water
- Heterogamy
- Absorption of Co2
- Protection of reproductive cells Formation of embryos
- Rhizoids for water absorption:

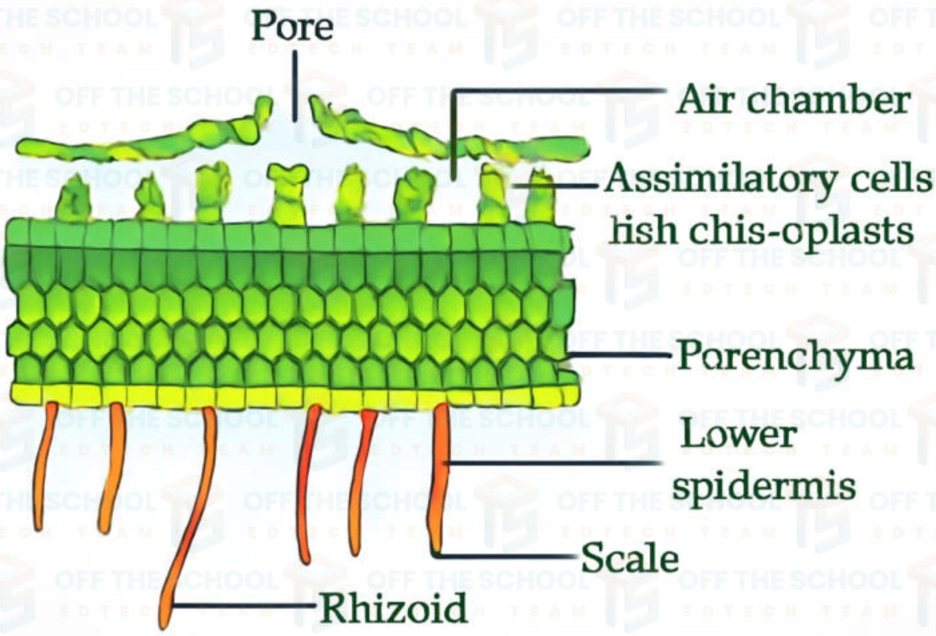
The study of Marchantia thallus and other bryophytes show that they have rhizoids for water absorption. These are long, filmentous extensions of the cell of the lower surface of the thallus. They greatly increase the surface for absorption of water from the soil.

Conservation of water:

To conserve water within the body thallus of all bryophytes is multilayered and very small surface area directly exposed to reduce the drying effects of the atmosphere. Moreover, the outer and uppermost layer of called cutin This is is is non-cellular layer of wax-like substance called cutin. This is very efficient in reducing the rate of evaporation.

Absorption of Co2:

Land plants need an efficient means for the exchange of gases with the environment in contrast . to aquatic plants which exchange gases dissolved in water. The upper surface of the thallus is provided with a number of aerating pores. Each pore leads inside into an air-chamber These chambers have filamentous branching photosynthetic cells. Air having CO_2 enters through the pores and CO_2 from air absorb by these photosynthetic cells and diffuses into the cytoplasm.



Heterogamy:
Heterogamy is the most successful kind of reproduction that has evalved in bryophytes. It is defined as production of two different trypes of eveles. One is male (motile) and the other is female (non-motile) full of atored food.

protection of reproductive cells:
The land environment requires special protection for the reproductive cells. In amphibious plants reproductive cells are very well protected. The male gamete (sperm) and female gamete (ovum) are produced in multicellular reproductive sex organs called antheridia and archegonia meetively. Moreover, these organs are also surrounded by hair like structures called paraphyses which help to prevent drying of the sex organs.

Formation of embryos:
Embryo formation in amphibious plant is of universal occurrence. The fertilized egg called oospore (zygote) is formed inside the archegonium. An embryo develops from the oospore as it divides still inside the protective coverings of the archegonia. Thus the coverings formed by the female organ protect the growing embryo from drying out and from mechanical injury.

Advantages of Bryophytes:
Bryophytes play important role in maintaining the ecosystem. They initiate soil formation, maintain soil moisture, and in recycling of nutrients in forest ecosystem. They are bio indicators of heavy metals in air pollution.

From ancient time in many parts of the world bryophytes used as medicines for example Liverworts (Marchantia) used for the treatment of liver disorder, another member of bryophytes Polytrichum used for curing diseases such as fever, homeostatic and traumatic injuries to pneumonia and lymphocytic leukemia. Insecticidal activities of bryophytes are also reported.

VASCULAR PLANTS:
Vascular plants characterized by the presence of vascular tissue. Vascular tissue is made up of Xylem and phloem. Vascular tissue has two important properties. First these tissues are concerned with translocation of water and nutrients throughout the plant body. Xylem carries mainly water and mind nut notte whereas Phloem carries mainly organic solutes in solution, such as sugar. Secondly plant bodies can be supported because Xylem contains lignified cells of great strength and rigidity. Another lignified tissue, sclerenchyma also develops in vascular plants and supplements the mechanical role of xylem. The dominant generation of living vascular plants is the sporophyte existing as free-living plant. The gametophyte is smaller in size, bear smaller sex organs. It may be monoecious (Bisexual) or dioecious (Unisexual), short lived, and degenerates once the sporophyte is established.
Vascular plants are categorized in two major groups. re categorized in two major (Psilopsida, Lycopsida Seedless
• Sphenopsida) and ferns.
• Sphenopsida) and terns. Seeded plants (Spermatophyte) ---- Gymnosperms and Angiosperms
• · Seeded plants (Spermant) Pteridophytes):
• Seedless Vascular Plant (Pteriaoped of mallies and ferns.
Fern allies:
The term " Fern allies" refers to a loose assemblage of spore production vascular plants that are classified in three group.
ular plants that are classined in throoms and tmesipteris.
• **(i)** Sub division Lycopsida e.g I shoradium and Selaginella **(ii)** Sub division Lycopsida e.g Lycopodium and Selaginella
• **(iii)** Sub division Sphenopsida e.g Equisetum

(i) Sub division Sphenopsida (Whisk ferns) living fossils:
(i) Sub division Psilopsida (Whisk fern of living vascular
This is the most primitive group of living vascular plants includes two genera Psilotum, Tmesipteris. The Psilopsida was considered as living fossile because of its apparent similarities to the fossil plant Rhynia. Psilotum is the only living vascular plant where leaves and roots are absent.

Life cycle of psilotum

The Psilopsida sporophytes are simple dichotomously branching small Plants. The lack true leaves but produce small out growths of stem (enations). The aerial stems are green and carry out photosynths of stem Sporangia develop at the tips of some of the acrial branches. Within the Sporangia meiosis produces haploid spores from spore mother cells. These
spores are morphologically similar therefore called homospory. Each spore spores are into gametophytes of Psilopsida which is also called prothallis. It has one or more dichotomous branches. It bears numerous unicellular This only is monocious . The male sex organ antheridium and female sex relan archegonium are produced near the growing apex.

Organization Lycopsida:

They are commonly called as club mosses most of which are tropical. They are the first plants to have evolved true roots. It is generally supposed that these roots arose from branches of the ancestral of algae that menetrated the soil and branched underground. The sporophyte consists of anchotomously branched shoots that bear simple spike like leaves. Sporangia are located in leaf axil either along normal portions of stem or condensed into cones, strobilus. These sporangia containing leaves are called sporophyll. The lycopodium are homosporous having one type of spores develop monoecious gametophytes that bears both male and female sex organs. Selaginella kraussiana is a common green house plant. Selaginella exhibits heterospory i.e., it produces spores of two distinct kinds large haploid spores germinate to produce female gametophytes, which archegonia development on it and small microspores which develop into male gametophytes, antheridium on it. These gametophytes are dioecious.

Sub division Sphenopsida (Horse tails):

Horsetail grow abundantly in the swamps Equisotum times Today the phylum consists of only one living genera Equisetum. It consists of a creeping underground rhizome that produces upright stems of a creeping indergrothetic. Small leaves are borne in sheeth like whorls at nodes in the stem. Sporangia are produced in small strobili borne terminally on either normal vegetative or specialized reproductive shoots. Following meiosis, spores are produced. Spores are unusual. The outer layer of spore wall form four sporopollenin thread called elators their movement assists in war dispersal. When a spore lands in a damp place, the elator coil up and spore germinates. A tiny gametophyte about the size of a few millimeters und to 3 centimeter is
produced.

Gametophyte is dorsiventral. Unicellular rhizoid are arised from the basal cells. Gametophyte is basically bisexual. It bears both male sex organ (antheridia) and female sex organ (archegonia).

Sub division Filicinophyta (Ferns):
Ferns are the most diverse and conspicuous of the spore - producing vascular plants. Most of these are found in tropical regions but many species grow abundantly in damp habitat in temperate regions. They range in size from small, delicate filmy ferns and floating aquatic to tall trees.

Ferns shows heteromorphic alternation of generation in which the sporophyte is dominant. The sporophyte generation possesses true roots, stems and leaves. Vascular tissue (Xylem and Phloem) present in the sporophyte. Leaves relatively large and called fronds. Spores produced in sporangia which are usually in clusters called sorus.

Gametophyte is reduced to a small, simple prothallus. The prothallus contains male reproductive organ antheridia and female reproductive organ archegonia. e.g. Dryopteris.

Evolution of the leaf:
The leaf is the most important organ of a green plant because of its photosynthetic activity. There are two types of leaves in vascular plants.
• (a) One veined leaf (microphyllous)
• (b) Many veined leaf (megaphyllous)

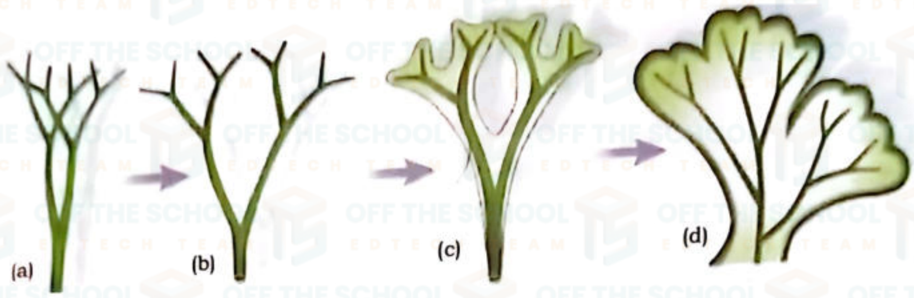
It is very interesting to trace the origin of leaf in the green plants. Evolution of One Veined Leaf: Enation Theory:

The evolution of one-veined leaf (microphyllous) can belexplained by assuming that a thorn like outgrowth (Enation) emerged on the surface of the naked stem. With an increase in size of the out growth, the vascular tissues were also formed for the supply of water, food and support to the leaf.

Reduction Theory:
Another possibility is that a single veined leaf originated by a reduction in size of the part of the leafless branching system of the primitive vascular plant. This is how the leaf of lycopodium (club mosses) and equisetum (horse tail) came into existence.
Evolution of Many Veined Leaf:

Webed Theory:
Many veined leaf (megaphyllous) originated much later. These are the evolutionary modifications of the forked branching system in the primitive plant. The first step in the evolution of this leaf was the restriction of forked

branches to a single plane. The branching system became flat . The next branches to a single plane filling the space between the branching and the branches to a single plane. The branching system became and the step in the evolution was filling the space between the even foot of a duck vascular tissues. The leaf so formed looked like the web foot of a duck vascular tissues. The leaf



3.3 Life cycle of a fern:

Life cycle of a fern: The life cycle of fern (Adiantum) or Dryopteris shows heteromorphic The life cycle of fern (Adiantum) or Dryoptens "Shown", "Showinant. All ferns alternation of generation in which sporophytic phase is dominant. All ferns the success in the I are homosporous, producing single types of spores.

Sporophytic Phase:

The sporophyte (2n) which is diploid, consists of adventitious roots, underground stem a rhizome and pinnately compound leaves.

ground stem a mizonic and philiates give rise to rounded bodies on its undersides which are called sori (singular sorus) .The sori are green but its underslues which are calloa own. The leaves which bears sori are called sporophylls. Each sorus is the group of sporangia. Sporangium consists of a stalk called sporangiophore and a biconvex capsule consisting of annulus and stomium. The annular cells are thickened whereas stomial cells are thin-walled. Within sporangium spore mother cells are present. Each spore mother cell divides by meiosis to from four haploid spores. The spores are liberated through stomium. Each spore on germination gives rise to miniature bisexual (monoecious) gametophyte called prothallus.

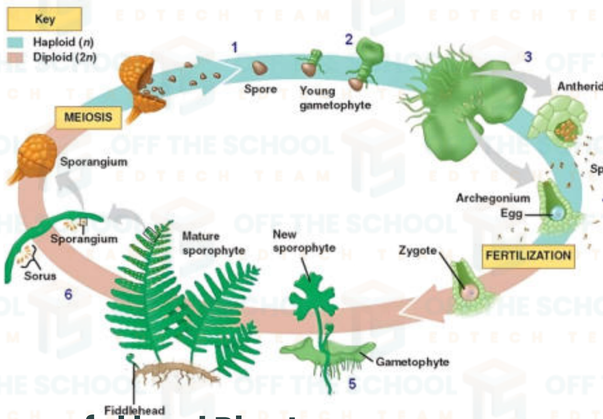
Gametophytic Phase:

The gametophyte of fern is called prothallus. The prothallus is independent autotrophic, heart shaped, dorsoventrally flattened lying prostrate on same wet substratum. It is fixed soil with the help of rhizoids which absorb water and nutrients. The prothallus is monoecious having archegonia and antheridia on the same prothallus

Archegonium consists of venter with an ovum and a neck and secrete malic acid by degenerating neck canal cells at maturity. Each antherdium produces a number of antheroziod (sperms). A number of sperms, by making chemotactic movement in water reach to the archegonium. Only one

Diversity Among Plants

spera fuses with ovum to form oospore (zygote) which is dilpoid. Young sporal fuses with ovum the oospore. In the meantime prother meantime prothallus sporophyte "develops" from" the "oospore." sporophyte in this way life cycle is completed.



Vascular Plants as successful land Plants:

The vascular plants have evolved a number of adaptations to the terrestrial environment that have enabled them to invade all the most inhositable land habitat. During this process they diverged sufficiently from mositable land nabitat. Damig tal adaptations of vascular plants (with a few minor exceptions)make them more successful land plants. • A protective layer of sterile jacket cells around the gametangia.

• Multicellular embroyo retained within the archegonia. ii.
• Cuticles on the aerial parts to prevent evaporation of water. iii.
• Cuticles on the acriar patts tand, dissolved minerals as well as IV . providej support.

Many other such adaptations, absent in the earliest vascular plants, appear in most advanced member of the division.

Importance of Seedless Vascular Plants:

Seedless plants have historically played a role in human life. Man gen many advantages including food, shelter, fuel and medicine and decoration,

They had been used as food e.g. Pteris, Ceratopteris and Marsilea Polypodium glycorrhiza is a part of diet due to sweetness of it's rhizome, and for treatment to certain diseases.

Coal, a major fuel source and contributor to global warming, was deposited by the seedless vascular plants of the carboniferous period.

The water ferns of genus Azolla harbor nitrogen fixing cyanobacteria and restore this important nutrient to aquatic habitats. The beautiful contr of ferns make them a favorite decorative plants.

The rhizome of Polypodium glycorrihiza (licorice fern) also have medicinal properties and is used as a remedy for sore throat.

SEED PLANTS: (SPERMATOPHYTES)

The seed plants are the most successful group of plants because they evolved a number of adaptations to a terrestrial habitat. These features which becomes the the female gametophyte and embryo within an ovule which becomes the seed. Microspores transported to the ovule as pollen grain through different sources.

Evolution of Seed:

Seeds are one of the important parts of plants, the presence of embryo inside the seed, necessary for the continuity of generation. Seeded Evolution of seed is one of the key recessful group of plant kingdom. Evolution of seed is one of the key reasons for the success of higher vascular plants.

There are three steps in the evolution of seed.

- **i.** Origin of heterospory i.e., two types of spores are produced. and megaspore grows into sperm forming gametophyte (male gametphyte) and megaspore grows into egg forming gametophyte (female
- **ii.** Development of integument, an enveloping structure for the protection in of megasporangia to form ovule. of megasporangia to form ovule.
- **iii.** Research portuigia to form ovule. gametophyte

The development of seed has given the vascular plants better adaptation to their environment.

Groups of Seeded Plants: (Spermatophytes)

- The seed plants have traditionally been divided into two groups

1. The Gymnospermae
2. The Angiospermae

The Gymnospermae:
They are the successful group of plants of worldwide distribution, acrounting for about one third of the world's Forest

General Characterstics of Gymnosperms:

- They have naked seeds because ovules are not covered by ovary i.e. ovary absent.
- They are trees or shrubs mostly ever green with needle like leaves with xerophytic adaptations.
- They are cold resistant and found in colder parts of the world.
- The tree is the sporophyte generation heterosporous producing two types of spores.

The best-known group of ggmnosperm is the conifers. In conifers ovules (later seeds) are located on the surface of specialized scale leaves called ovuleiferous scales .These are arranged in cones. scale leaves called ovuleiferous scales .These are tree Male and female cones are produced on the same tree.

- Most conifers bear their micro and megasporangiate structure in cones · Most conifers

bear their micro and micro al megaspore, develops

- Gametophyte generation reduced , the functional megaspore develops

Gametophyte
generation reauced , the fametich is permanently lodged within megasporangium (Ovule)

• Microsporangia (Containing pollen) are borne on microphylls (Fertile
• Vascular tissue containing Xylem tracheids and Phloem sieve cells · Vascular tissue containing bableam companion cells are absent. Vascular tissue containing Aylene companion cells are absent. while Xylem vessels and phloem companion five archegonia ea
• Female gametophyte consists of two to five archegonia each having single ovum.
• Each microspore develops into another miniature male gametophyte consisting of stalk nucleus tube nucleus , two male gametes and two consisting of stalk nucleus tube nucleus , two male gametes and two consisting of stails interest within an elongated pollen tube.
• Pollen grains of conifers are wind dispersed.
• Fertilization takes place within sporophyte.
• Fertilization takes pace oorangium gives rise to seed.
• Seed undergoes epigeal germination to form new sporophyte plant. e.g. · Seed undergoes epigeal germination to form new sporophyte plant.
• Seed undergoes epigear get larch, cedar redwood, Yew.
Conifers are the most diverse and wide spread of the living non llowering seed plants. The specificantly plants many of which are important soft wood timber tree. ifers are the most diverse and wide spread of the Inting non nowering of the

Uses of Gymnosperms:

• · Conifers are commercially important, conifer wood is valuable commercial resource world wides. Pines, spruce firs and other conifers are a major source of softwood timber used in house construction, the timber can be pulped to make paper or compressed to make chipboard, the basic material for modern furniture.
• In addition, Pines are a source of resins, turpentine and pine oil.
• · The plants are widely used as ornamentals and largely spotted in gardens and parks e.g Cycas, Thuja, Juniperus, Ginkgo.

• The famous dry fruits locally called chilghoza (Pinus gerardiano) belongs to pine tree.
• · Ephedrine a drug obtained from Ephedra is used in various r espiratory disorder particularly in asthma.

The Angiosperms:

The flowering Plants, or angiosperms have their seeds enclosed in fruit because ovules are covered by ovary They are dominate vascular to polar regions, included to a great range of habitats from the tropics to polar regions, including frets and salt water.
The smallest plants are free floating aquatic duck weeds which is highly reduced and not even differentiated into leaf and stem. The tallest flowering plant is the mountain ash tree,
One of the most important features of angiosperm a part from the enclosed seed is the miss important reatures of anglosperm a part from the cones.
Flowers are unique to angiosperms and have become adapted to attracts insects or other pollinators, for example by the development of large brightly coloured petals, nectaries and scent glands brightly coloured petals, nectaries and scent glands.

Life cycle of flowering Plants:

Flowering plants show heteromorphic alternation of generation . The gametophyte hapid gametophyte alternates with diploid sporophyte . The gametophyte . The gametophyte happid gametophyte alternates with diploid sporophyte. haploid gametophyte alternates with diplom sporophyte.

Sporophyte generation:

Asexual reproduction in Angiosperm:

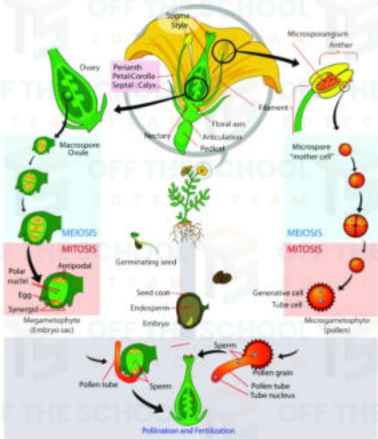
The main plant body is sporophyte that consist of root, Stem, leaves and flowers. Root, stem and leaves are considered as vegetative parts while and flowers. Root, stem and leaves are considered reproductive shoot flower is the reproductive part. Flower is a compressed reproductive shoot with four whorls of modified (floral) leaves called sepals, petals, petals, with four whorls of modified. Microsporophyll (carpel), is microsporophyll (stamen) and megasporophyll (carporophyll (carpel) is female reproductive structure.

Microsporophylls (Stamens):

Microspores which later develops into pollen grains are produced by microsporophyll or stamens, each stamen consist of sac like anther and a stalk called filament, usually the anther have four chambers called pollen stalk called filament, usualiy the anther have four chambers called pollen stalk called filament, u sac or microsporangia, contain numerous microspores mother cells. sac or microsporangia, contain numerous microspores by meiosis.Each microspore mother cell produces four microspores by meiosis so Each microspore mother cell produces four loss of a loss grains and disperse from here for pollination.

Megasporophylls (Carpels):

Megasporophyll now called carpel produce ovum in ovary. Each carpel consist of three parts, stigma, style and ovary, within overy one or more ovules (megasporangium) develop. The main body of ovule is called nucellus which is protected by two layers of protective sheaths of cell callege integuments. A small pore is present on it, the micropyle, with these in contains one megaspore mother cell which produce four megaspores on maturation. Out of four megaspores only one megaspore will be functional to develop into female gametophyte. The female gametophyte develops inside ovule called embryo sac.



Development of female gametophyte:

Megasporation (Ovule) within ovary contains mega spore mother Thee of them disintegrate and only one survive haploid megaspores. Three of them disintegrate and only one survive, which megaspores. Three of them disintegrate and only one mitten division develop into seven-cell (eight one survive, which after successive mitotic division develop into seven-cell (eight and only one mass morther mitotic division develop into seven-cell (eight nucleation) structures successor server successores. Imale gametophyte or enbryo sac. Out of seven three estiled chlaza and called antipodal cells, three cells, three cells found toward called egg apparatus, two of which termed stound themator microphyle colled called poly large cell is egg cell and one central sell mich media microporities and one called polaratus, two of which termed spacersion celler and one called polamus, atter fusion and one central cell with two nucle sametes All of ocells except secondary ralled secondary nation of the climitive nuclei. All fo octla except secondary ralled secondary nationalized minimelian in stucture of order is termecoeders of nusion called secondary nuclein of deliver of delevise of delevise of delevise of delevise of de condition of declein of decline of decliner of decli structure of ovule is ept secondary nuclei are haploid. This multies and Section of ovule is termed embryo sac which form female gametophyte

Double fertilization:

After pollination, pollen grains transfer and settle on the stigma of , The pollen tube grows through the tissue of the stigme on her the enters the overy with tube grows through the tissue of the stigma and style than enters the overy with tube nucleus at its stigma and style the reaches on ovule, it enters the tise when the signal and the stigma and the stigma and the pollen sperm cells into the female gametoplyte and then disclarges the partner of the partnerges the por the partnerges the perm the egg cell to form zygote (2n) which micropyte and One bacharges the oblies barges the bollen the egg cell to form zygote (2n) which firther desche one sperm fertilizes sporophyte. By the time fertilization occurs the daviders into an embryo female gametophyte have combined to form a the two pps into an embles of the which the second sperm unites to form a diploid form a diploid foolon nucleus of the lost tof the lost tof the succles with which the second sperm unites to form a triploid nul fleus s mucless nucleus nucleus nucleus nucleus nucleus nucleus miss nucleus undergoes a series of division and a triploid cell, called endosperm is formed. The endosperms function in the food for the embryo.

Food storage organ provide food to embryo at the time of germination. Double fertilization is a special type of fertilization which occurs only in angiospermic plants. During this process a sperm fuses with the ovum in form oospore. The other sperm fuses with secondary nucleus to form triploid endosperm nucleus. endosperm nucleus.

After fertilization the ovule matures and develop into seed while ovary develops into the fruit. The fruit not only helps to protects the seeds from desiccation during the early development, but often also facilitates their dispersal by various means.

Angiosperms as successful group of land Plants:

The life cycle of angiosperm shows that angiosperms are so well adapted to life on land. Their major advantage over other plants is related to their reproduction. Here they are better adapted in three important ways.

1. The gametophyte generation is very reduced. It is always protected inside sporophyte tissue, on which it is totally dependent .In mosses and liverwort where the gametophyte is conspicuous ,while in seedless vascular plants it is a free-living prothallus. The gametophyte is susceptible to drying out. 2. Fertilization is not dependent on water as it is in other plant groups, where sperms swim to the ovum . The male gametes of seed plants are non-motile and are carried within pollen grains that are suited for dispersal by wind, insects, water and animals. Finally transfer of male gametes after pollination takes place by means of pollen tubes. The ova being enclosed within ovules.

Conifers and flowering plants produce seeds. Development of seeds is Conifers and flowering plants produce seeds. In their contents on the parents sporophyte

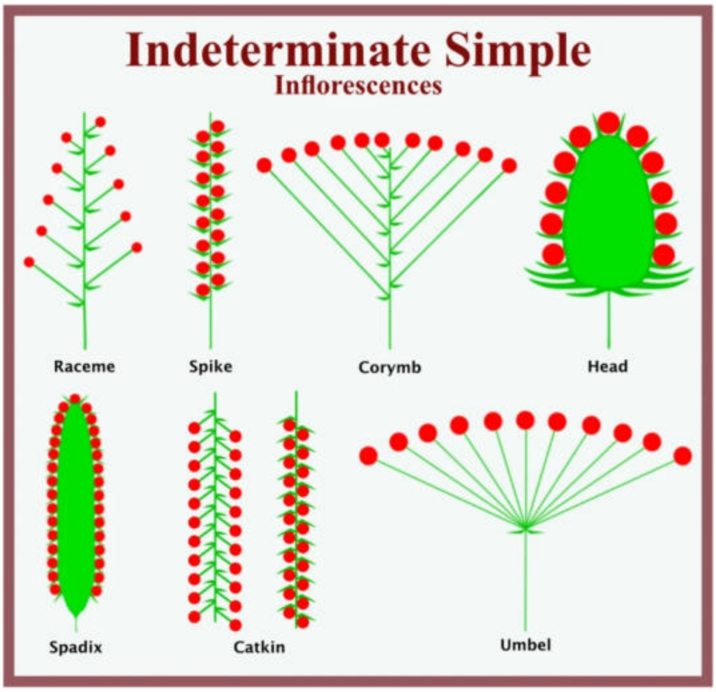
Inflorescence:

It is mode of branching of floral axis having a group of flowers on it may be defined as arrangement of group of flowers on a floral axis. There are may be defined as arrangement of group or nowe. In recemose inflorescence, main axis called pendunes continues to grow. The flower develop in acropetal succession and opening of flower is centripetal. Whereas in acropetal succession and opening of flower is centrifugal. cymose innorescence, no opening of flower is centrifugal.

Kinds of recemose inflorescence:

Peduncle elongated:

Receme is an inflorescence in which flowers are pedicellate and bisexual e.g., Gold mohar whereas in spike, the flowers are sessile and bisexual (amaranthus) Catlein is another kind of inflorescence in which flowers are sessile and unisexual (Mulberry) whereas in spadix, flowers are covered over by one or many large bracts called



peduncle shortened:
Corymb is an inflorescence in which flowers have pedicels of unequal length lowers having large pedicels and upper ones having small length lowers having large of same length and arise from common point, e & Iberis) when pedicels are of same length and arise from common point, (e.g. Ibers)

the typed flattened:
Head (capitulum) flattened peduncle has a mass of small sessile Bowers (florets) with one or more whorls of bracts at the base forming an involuce. The floreds are commonly of two kinds namely ray florets involucre. The florets are commonly of two know namely as a sunflower, zinnia, marigold etc.

Spikelet Inflorescence:
It is kind of recemose inflorescence. There are three bracts at its base called glumes. The lower two without flowers are called empty lemma, there is small bracteoles called Palea. Flowers, covered by their respective lemma is small bractedies called Patea. Flowers, covered by and family Poaceae.

Kinds of cymose inflorescence: Uniporous (Monochasial) cyme:
Main axis soon ends into a flower and produces only one lateral branch at a time ending in a flower. The succeeding branches again follow the same mode of producing flower. If the succeeding branches are follow the same mode of producing howers are produced on same side, it produced on alternate sides, "It is called bees are produced on same side, it is called helicoid cyme (sundew).

Biparous (dichasial) Cyme:
Main axis soon terminate into a flower and produces two flower . This mode is followed by each succeeding flower (Pink night - jasmine). mode is followed by each succeeding flower (Pink night - jasmine). The is followed by each succeeding fl
8.4.7 Significance/benefits of anglosperms for human. They haveng Angiosperms are very important to human. They have number of uses
as food; specially cereals from the grass family. Among most economical as food; specially cereals from the grass family. Allong moot securitiedly important grains are corn (Zeamays), barley (Hordium), oats (Avena), wheat important grains are cor (Triticum) and rice (Aryza).

Vegetables:
There is a variety of vegetables, different in forms and nutritional There is a variety of vegetables, differe of their parts the flowers, shoots, content, and are grown for one or more of their parts tomato, eggplant and chill, leaves or the underground parts. E.g., Potato, tomato, egginant and tohili, leaves or the underground parts. E.g., Potato, comes and beans (Fabacine), Mann pepper from the potation from the squash family (Cucurbitaceee), pumpkin, melon and gourd from the squash from mustard from mustard family (Brassicaceae).

Fruits:
A large number of fruits belongs to rose family (Rosaceae), e.g. Almonds, apples, apricots, cherries, peaches, pears, raspberries and Almonds, apples, apricots, cherrites, peaces, bananas from family Mucaceae and Papaya from family Caricaceae.

Medical / Pharmaceutical plants:
Flowering plants have great pharmaceutical and medical value. Citrus Flowering plants have great prin from bark of willows, marcotics such as opium from opium poppy, quinone from cinchona. Arq-e-gulab extracted as optum from opium poppy quas eye drop and for soothing. The pulp of amaltas (Cassia fistula) acts as purgative. Atropa belladonna of family Solanaceae, Solanum nigrum locally called mako used as pain reliever and anti-inflammatory. Ocimum tenuiflorum commonly called tulsi used in many herbal medicines to help treat asthma, bronchitis, cold and flu. Glycyrhiza glabra locally called mulhethi traditionally used to treat many diseases particularly in respiratory disorders.
Angiosperm compounds that are highly toxic to humans have proved to be effective in the treatment of certain forms of cancer, such as acute leukemia (vincristine from the Madagascar periwint).